PROBLEM 3

SIMULATION VS PRACTICAL DEVELOPMENT

Technical proposal

Objective. Choose a development approach for an RC aircraft being prepared for a national aeromodelling competition, with limited time, money and resources.

Recommended approach: hybrid and test-led. Use calculations and simulation to screen designs, build a low-cost prototype early, then compare measurements with predictions and refine the design. Simulation enables quick comparisons, but cannot fully capture build errors, battery voltage sag, vibration, control friction, surface finish or launch conditions.

Scope note. This is a proposal based on the supplied assignment brief, not evidence that an aircraft has been built or tested. Numerical predictions remain preliminary until supported by component datasheets and physical measurements.

1. Comparison of development approaches

Criterion	Simulation-led	Prototype-led	Hybrid recommendation
Cost	Software/setup may take time or money; simulation can prevent some failed builds.	Repeated builds can waste materials.	Use available tools and inexpensive early materials.
Time	Parameter sweeps are fast once a model exists; setup can be slow.	Fast to start, but redesign needs fabrication or repair.	Simulate high-impact choices and build early.
Accuracy	Depends on assumptions and model fidelity.	Tests use real hardware but may be noisy.	Compare predictions with repeatable measurements.
Iteration	Geometry changes and reruns are quick.	Iteration is limited by cutting, assembly and curing.	Use simulation to screen options; tests to validate.
Failure detection	Finds trends and modeled geometry or stress issues.	Finds real assembly, electrical and handling issues.	Use both; complete ground checks before flight.
Final performance	May optimize a model while missing real-world effects.	Can work, but trial-and-error may miss improvements.	Seek agreement between model and measured behaviour.

2. Proposed workflow

1. Define mission, competition rules, site constraints, safety and success metrics.

2. Estimate mass budget, wing loading, stall speed, thrust requirement and centre-of-gravity (CG) range.

3. Create CAD geometry; check dimensions, component packaging, control travel and manufacturability.

4. Use suitable tools: XFLR5 for preliminary aerodynamic trends; Fusion 360/SolidWorks for geometry; structural FEA only with credible loads and material properties.

5. Build a simple, inexpensive and repairable prototype.

6. Inspect structure/alignment; measure mass and CG; verify control direction/travel, radio range/failsafe, motor rotation, propeller security, current and temperature.

7. Compare measurements with predictions and investigate causes of disagreement.

8. Only after checks pass, conduct controlled flight tests at a suitable legal site with experienced supervision; change one variable at a time.

9. Record results, repair/refine, repeat, and freeze the design with contingency time.

Development flow: Requirements → Hand calculations → CAD/simulation → Prototype build → Ground inspection and power checks → Compare predictions with measurements → Refine model/design → Controlled flight test → Review/design freeze. If a safety check fails, return to diagnosis; do not proceed to flight.

3. Parameters to simulate and verify physically

Simulation candidates

• Lift/polar trends and estimated stall speed for candidate airfoils.
• Lift distribution and aerodynamic-centre trends.
• Stability trends and sensitivity to CG.
• Structural deflection/stress for a clearly stated load case.

Physical verification

• Finished mass and CG.
• Static thrust, current and voltage with the actual power system.
• Control direction and travel; radio range and failsafe.
• Alignment, vibration and structural integrity.
• Observed glide/flight behaviour during controlled tests.

Simulation results are estimates, not guarantees. Choose parameters that can be measured or checked consistently in the real aircraft.

4. Validating simulation

• Match geometry, mass, CG, battery condition and test conditions as closely as possible.
• Measure repeatable quantities: mass, CG, dimensions, voltage, current, static thrust, control deflections and temperature.
• Percentage error = |measured – predicted| / |measured| × 100%. Use absolute error if the measured value is near zero.
• Repeat measurements, record instrument resolution and avoid tuning a model to a single noisy result.
• If results disagree, check actual geometry, alignment, surface finish, propeller, voltage sag, sensor calibration and assumptions.
• Record model version, test setup, result and design change. Check the revised model against measurements not used to tune it.

5. If simulation predicts flight but the prototype cannot take off

1. Stop repeated attempts. Remove the propeller for radio/control checks and secure the aircraft for powered tests.
2. Check battery state, connector polarity, motor rotation, ESC arming/calibration, propeller orientation/condition and motor–propeller compatibility.
3. Measure current and voltage with suitable equipment; compare with manufacturer limits. Look for battery sag, hot components or loose connectors.
4. Check finished mass and CG against the design; excess mass or incorrect CG can prevent safe launch.
5. Check wing incidence, tail alignment, wing twist/warping, binding linkages and correct control directions.
6. Reconsider assumptions: actual airfoil/build shape, surface roughness, low airspeed, insufficient thrust or excessive wing loading.
7. Check launch method, surface and wind. Do not blindly increase power or control throws.
8. Change one variable at a time, log results and seek experienced supervision before further flight.

6. Four-week schedule

Week	Tasks	Gate / contingency
1	Requirements, calculations, mass budget, preliminary CAD and component shortlist.	Check assumptions and part availability; identify backups.
2	Aerodynamic screening, layout, material selection and prototype build.	Build a simple prototype; avoid cosmetic work.
3	Assembly, CG/control/radio checks, failsafe and restrained power tests.	Reserve 1–2 days for repairs/delays; no flight if checks fail.
4	Controlled flight tests, refinement, documentation and review.	Keep final 2 days as contingency; freeze only after safe, repeatable results.

Safety and source note

The supplied brief establishes the required tasks but provides no exact aircraft geometry or measured results. Use manufacturer datasheets and actual measurements before making firm claims. LiPo batteries and rotating propellers can cause serious injury or fire; follow safe handling practices and use experienced supervision.